

Multiple Choice (10 marks)

1. The red glow in the neon tube of an advertising sign is a result of
 - (a) Refraction
 - (b) de-excitation
 - (c) Chemiluminescence
 - (d) Thermal radiation
 - (e) Phosphorescence
2. A coil of 60 ohms inductive reactance in series with a resistance of 25 ohms is connected to an a-c line of 130 volts. What will be the current through the circuit?
 - (a) 2.5 amp
 - (b) 0.5 amp
 - (c) 2.8 amp
 - (d) 3.0 amp
 - (e) 1.0 amp
3. If light of wavelength 635 nm causes a certain amount of Rayleigh scattering, light of what wavelength will give exactly 10 times as much scattering?
 - (a) 700 nm
 - (b) 450 nm
 - (c) 750 nm
 - (d) 300 nm
 - (e) 357 nm
4. At what velocity is the mass of a particle twice its rest mass?
 - (a) 3.2×10^{10} cm/sec
 - (b) 3.0×10^{10} cm/sec
 - (c) 2.8×10^{10} cm/sec
 - (d) 2.4×10^{10} cm/sec
 - (e) 1.8×10^{10} cm/sec
5. The pressure at the bottom of a pond does NOT depend on the
 - (a) salinity of the water.
 - (b) proximity to the equator.

- (c) presence of fish in the pond.
- (d) type of pond vegetation.
- (e) acceleration due to gravity.

True / False (10 marks)

Answer True or False

6. True or False: The Mars Exploration Rover Spirit tilts towards the north because it is located in the southern hemisphere during winter.
7. True or False: The reflectance for light of normal incidence at a water-glass boundary is 0.50% for crown glass and 1.00% for flint glass.
8. True or False: The frequency of the next higher harmonic for a closed organ pipe with a fundamental frequency of C (131 Hz) is 196 Hz.
9. True or False: It is possible for an object to have a net acceleration of 0 m/s^2 when acted upon by multiple unbalanced forces of equal magnitude.
10. True or False: The impedance of a 1-henry inductor at angular frequencies of 100, 1000, and 10,000 rad/sec is 500 ohms, 5000 ohms, and 50 ohms, respectively.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A 5-cm plane mirror can reflect the entire image of a tree when positioned 30 cm from an observer, 100 m away from the tree. Calculate the height of the tree and discuss the underlying principles of reflection involved in this scenario.
12. Discuss the relationship between operating temperature and efficiency in heat engines, such as jet engines. How do high temperatures enhance their performance?

End of Paper